Arithmetic of Recovery — V5.33

Godavari / Nashik Decision Dashboard — Model Analysis and Supporting References

1. What the dashboard is

This is a self-contained HTML/JavaScript "decision dashboard" built around the Ramkund bathing ghat on the Godavari River in Nashik, India, positioned as pilot decision support ahead of the Simhastha Kumbh Mela 2027 ("Simhastha 2027"). It combines satellite remote sensing, a simple physical/statistical water-quality model, weather data, and a cost-optimisation layer into one interactive tool. The dashboard's own framing is explicit that it is a **modelled planning aid, not a bathing-safety certification system** — several acceptance gates (microbial validation, forecast back-testing, causal source attribution, intervention field trials) are marked "OPEN" in the underlying code.

2. Core components of the model

Satellite evidence layer. Uses optical index proxies computed from Sentinel-2 imagery (via Google Earth Engine) — a turbidity index (NDTI-style), a chlorophyll index (NDCI-style), a floating-matter / algae index, and a modified water index (MNDWI) to isolate open water. These are relative, unitless optical proxies rather than calibrated concentration measurements.

Upstream-vs-local comparison. The bathing reach (Ramkund) is compared against an upstream "control" reach (near Gangapur/Someshwar) and a downstream recovery reach, so a given day's signal is read as a contrast rather than an absolute value.

Recovery half-life estimate. The dashboard's headline statistic — a disturbance loses about half its excess signal every ~1.55 days — is derived from fitting an exponential-decay / pulse model ($E(t) = \Sigma A_i \cdot 2^{-(t-t_i)/T_{half}}$) to the archive of detected excursions across a stated ~385-scene satellite archive.

Source-separation / mass-balance model. A simple tank (mass-balance) model, $dC/dt = (Q/V)(C_{in} - C) + (L_{bath} + L_{drain} + L_{runoff})/V - kC$, apportions the observed anomaly across bathing load, drain discharge, runoff and upstream contributions. The dashboard is explicit that these source probabilities are "diagnostic, not causal" until event-study, placebo and drain-tracing tests are run.

Forecast & risk module. Projects the state forward 6–72 hours using the fitted decay/mass-balance parameters plus live weather (rainfall) pulled from the Open-Meteo API, and reports a probability of remaining above a provisional threshold (κ).

Confidence score. A composite 0–100 score built from five weighted sub-scores (data coverage, sensor/pixel health, calibration quality, source-attribution certainty, data freshness); the code flags this confidence as intentionally capped while causal attribution and field calibration remain open.

Economic / intervention optimisation. A brute-force grid search over three intervention levers (upstream interception, local control, flushing/turnover) selects a spending plan that minimises expected decision loss plus a tail-risk (CVaR) penalty, subject to a user-set budget.

Auxiliary layers. A live-orbit "globe" view (via satellite.js / N2YO / CelesTrak two-line elements) visualising the relevant Earth-observation satellites; an interactive Leaflet map of the study reaches; and PDF/Excel export and audit-log functions built into the page itself.

3. What the underlying data actually is

It is important to note that the numeric outputs are **modelled/demo values**, not verified field measurements: the model card embedded in the file's own export function labels several core parameters as "assumed/demo," "modelled from crowd until event-study coefficient fitted," or "modelled from rainfall until runoff calibration." Cost inputs in the economics module are described as "user-entered assumptions" that must be replaced by engineering estimates before any financial claim is made. The 75% "evidence confidence" badge shown on the conclusion screen is explicitly scoped to the 385-scene archive, the 1.55-day half-life estimate and an internal permutation check — the dashboard states this is **not** a formal accuracy score or safety certification.

In short: the dashboard is best read as a well-engineered *analytical framework and pilot proposal* (remote sensing → recovery-time estimate → source diagnostics → timing/budget recommendation → verification loop) rather than as a peer-reviewed water-quality study with field-validated numbers.

4. Positioning / stated purpose

The dashboard frames its contribution as a "recovery-aware decision layer" over satellite evidence: measure what changed, estimate how fast it clears, distinguish local from background conditions, and time interventions accordingly — intended to help government/municipal teams decide where to look first, when to act, and when not to spend limited resources on action that is too early or too late, ahead of the large-scale Simhastha 2027 pilgrimage event.

5. Research papers cited in the dashboard

The dashboard includes its own "Research papers & public references" panel (and an equivalent list in its built-in PDF-export function) to document the methodological basis for its remote-sensing indices and data platforms. These are reproduced below, extracted directly from the file.

Gorelick, N. et al. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202, 18–27. DOI: 10.1016/j.rse.2017.06.031.

Relevance: Basis for using Google Earth Engine as the scalable satellite/geospatial processing platform.

Mishra, S. & Mishra, D. R. (2012). Normalized difference chlorophyll index: a novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Remote Sensing of Environment*, 117, 394–406. DOI: 10.1016/j.rse.2011.10.016.

Relevance: Methodological basis for the red-edge chlorophyll (NDCI-style) proxy used in turbid productive waters.

Hu, C. (2009). A novel ocean color index to detect floating algae in the global oceans. *Remote Sensing of Environment*, 113(10), 2118–2129. DOI: 10.1016/j.rse.2009.05.012.

Relevance: Basis for the floating-matter / algae-sensitive spectral index family.

Lacaux, J.P. et al. (2007). Classification of ponds from high-spatial resolution remote sensing: application to Rift Valley Fever epidemics in Senegal (source of the NDTI turbidity-discrimination approach cited by the dashboard).

Relevance: Reference basis for the normalized difference turbidity index (NDTI) used as a relative optical turbidity indicator.

Xu, H. (2006). Modification of normalised difference water index (NDWI) to enhance open water features in remotely sensed imagery. *International Journal of Remote Sensing*.

Relevance: Reference basis for the Modified Normalized Difference Water Index (MNDWI), used for open-water discrimination.

6. Public data / documentation sources cited

Alongside the peer-reviewed references, the dashboard cites the following public data platforms and documentation as the source of its live/near-live inputs:

- Copernicus Data Space Ecosystem — STAC API documentation (Sentinel-2 L2A scene metadata/acquisition retrieval).
- ESA Copernicus Sentinel-2 mission, product and sensor documentation.
- Open-Meteo API documentation (live temperature and rainfall/weather forecast data).
- Google Earth Engine platform documentation (incl. its commercial-transition guide).

7. Additional public links referenced in the page (satellite / imagery sources)

The file also embeds a hidden "scaleSources" list of external links, presumably documenting where a future/production version could source higher-resolution Earth-observation imagery and administrative approvals:

Source	URL
Divisional Commissioner Office, Nashik — Administrative Approvals	https://divcomnashik.maharashtra.gov.in/en/administrative-approvals/
ISRO — Earth Observation Satellites	https://www.isro.gov.in/ISRO_EN/EarthObservationSatellites.html
ISRO — RESOURCESAT-2A	https://www.isro.gov.in/RESOURCESAT_2A.html
ISRO — EOS-06 (PSLV-C54 mission page)	https://www.isro.gov.in/mission_PSLV_C54.html
Google Earth Engine — commercial transition guide	https://developers.google.com/earth-engine/guides/transition_to_commercial
NASA Commercial Smallsat Data Acquisition (CSDA) — Pixxel	https://science.nasa.gov/earth-science/csda/csda-vendor-pixxel/
Airbus — Pléiades Neo optical/radar satellite imagery	https://space-solutions.airbus.com/imagery/our-optical-and-radar-satellite-imagery/pleiades-neo/
Planet — SkySat imagery documentation	https://docs.planet.com/data/imagery/skysat/

Note on sourcing: this list was extracted directly from the dashboard's own HTML/JavaScript source (its dedicated "Research papers & public references" panel, its built-in PDF-export routine, and a hidden auxiliary link block). No outside literature search was performed to independently verify these citations or to check whether the dashboard's numeric outputs are actually supported by them — that would be a reasonable next step before relying on this dashboard operationally.